Note

Lipids yield ~9 kcal of energy per gram, while carbohydrates and proteins yield only ~4 kcal of energy per gram.

6.2. Fatty acids

Fatty acids are the building blocks of hydrolysable lipids.

6.2.1 Classification of Fatty acids:

A simple fatty acid consists of a long linear hydrocarbon chain that may be saturated as in palmitic acid or it may have one or more double bonds as in linoleic acid. Few fatty acids like Arachidonic acid and Docosahexaenoic acid (DHA) contain more than three double bonds. Each fatty acid differs from the other, primarily in chain length, number and position of their double bonds. They are often symbolized by the number of carbon atoms in the hydrocarbon chain and number and position of the double bonds. Palmitic acid (Figure 6.3) is symbolized as 16:0 and oleic acid (Fig.6.3) as 18:1(Δ^9), Δ indicates the position of the double bond.

There are many kinds of fatty acids, isolated from various plant and animal lipids. Fatty acids with odd number of carbon atoms are significantly present in many marine organisms.

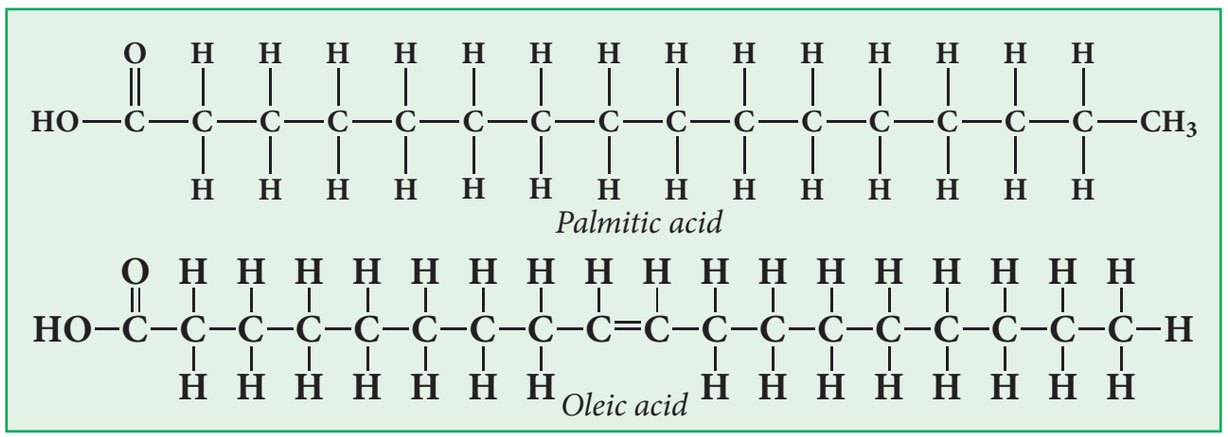

Figure 6.3 Structure of palmitic acid (saturated) and oleic acid (unsaturated)

Saturated Fatty acid

The general formula of saturated fatty acid is $C_nH_{2n+1}COOH$. If all carbon-carbon bonds in the hydrocarbon chain of the fatty acid are single covalent bonds, then the

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